



Diego Gerardo Sánchez Moreno

Robotics & Embedded Systems

Contact

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Languages

Spanish: Native

English: B2

Core Skills

ROS 2, micro-ROS, PID/PI

OpenCV, YOLOv8

ESP32, Jetson Nano

MATLAB, VHDL, C++

Supabase, PostgreSQL (RLS)

Next.js, FastAPI, Docker

Diego Gerardo Sánchez Moreno

Robotics & Embedded Systems Engineer

Co-founder & Project Director at Autónoma (AutoNomaMX), with three POS systems in production for Mexican businesses. Background in soft robotics control, embedded systems, and perception for mobile robots.

Education

B.S. in Robotics and Digital Systems Engineering

Tecnológico de Monterrey, Campus Querétaro

2022 – 2026

Professional license (Cédula Profesional) 15796458 — SEP, Mexico, 2026

Research & Engineering Experience

Autónoma (AutoNomaMX) — Co-founder & Project Director

Querétaro — Jan 2026 – Present

- Shipped three offline-first POS systems in production with real clients (a restaurant, a coffee shop, and an optical shop) on a multi-tenant PostgreSQL/Supabase backend with Row-Level Security.
- Built a SaaS analytics platform with an AI assistant mediated by a semantic layer; CI/CD with self-hosted runners on production hardware (Raspberry Pi, Ubuntu) with smoke tests.
- Directed Phase 1 of the optical-shop project under PMBOK: charter, WBS, CPM, delivery with signed acceptance, client training, and closure with \$0 cost deviation.
- Integrated POS hardware: ESC/POS thermal printers and Code128 barcode labels (Brother DK-1204).

Hybrid Soft Robotics Lab (Tec de Monterrey, Qro.)

Soft Robotics Control Platform — 2024 – Present

- Built a bi-stable pneumatic driver with closed-loop pressure/vacuum control, safety interlocks, and real-time telemetry.
- Implemented micro-ROS on ESP32 (UART) with pressure sensing over I2C; delivered a Python SDK + GUI and MATLAB-based analysis.

Instituto Asunción de Querétaro — Web Developer

2026 – Present

- Develop and maintain the institution's official website.

Patio Cinco (Querétaro) — Production Platform

Product/Systems Engineer — 2025 – 2026

- Delivered a production system for memberships, billing, and customer operations with persistent fiscal data.

Selected Projects

PuzzleBot Autonomous Mobile Robot (ROS 2)